Hydrogen Supply Chain Optimization

Final presentation

Mahammad Valiyev

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- MPC workflow components
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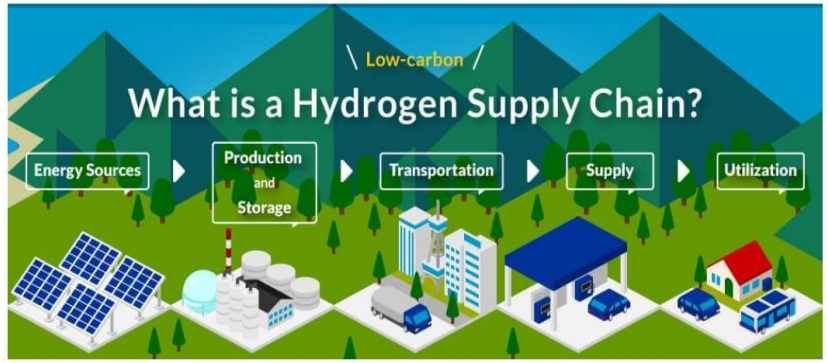
1. Introduction

Introduction

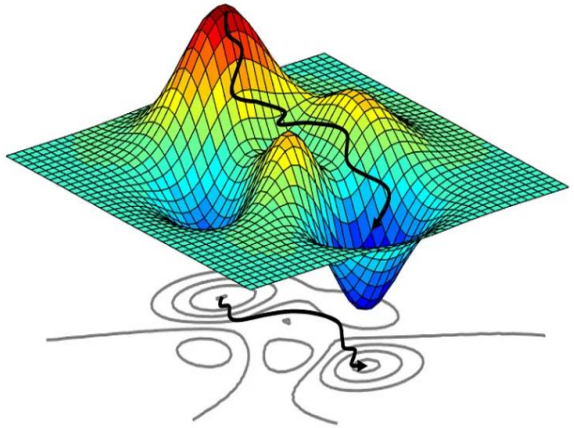
1. What is a supply chain?

- Flow of materials, energy, information across production, storage, and delivery
- Integration of suppliers, processes, logistics from source to end-user



2. Hydrogen supply chain

- Energy sources, hydrogen production
- Storage, distribution, export, utilization



3. Need for optimizing the hydrogen supply chain

- High production/storage cost, variable renewables, dynamic demand
- Improve efficiency, minimize emissions, ensure reliability, reduce waste

Solution Strategies for Hydrogen Supply Chain Optimization

1. Overview of approaches

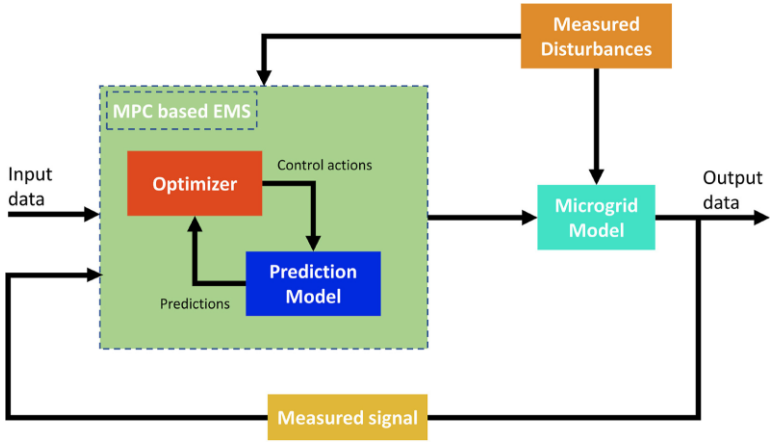
- **Rule-based**: predefined logic for operation
- **Heuristic methods**: empirical decision-making
- **Optimization-based**: mathematical cost optimization

2. Simplistic benchmark approach for study

- **Rule-based control**: Operates on fixed thresholds and logic
- **Reactive adjustments**: no forecasting, responds to current state
- **No optimization**: simple logic, easy to implement

3. Optimization-based approaches: MPC, MILP

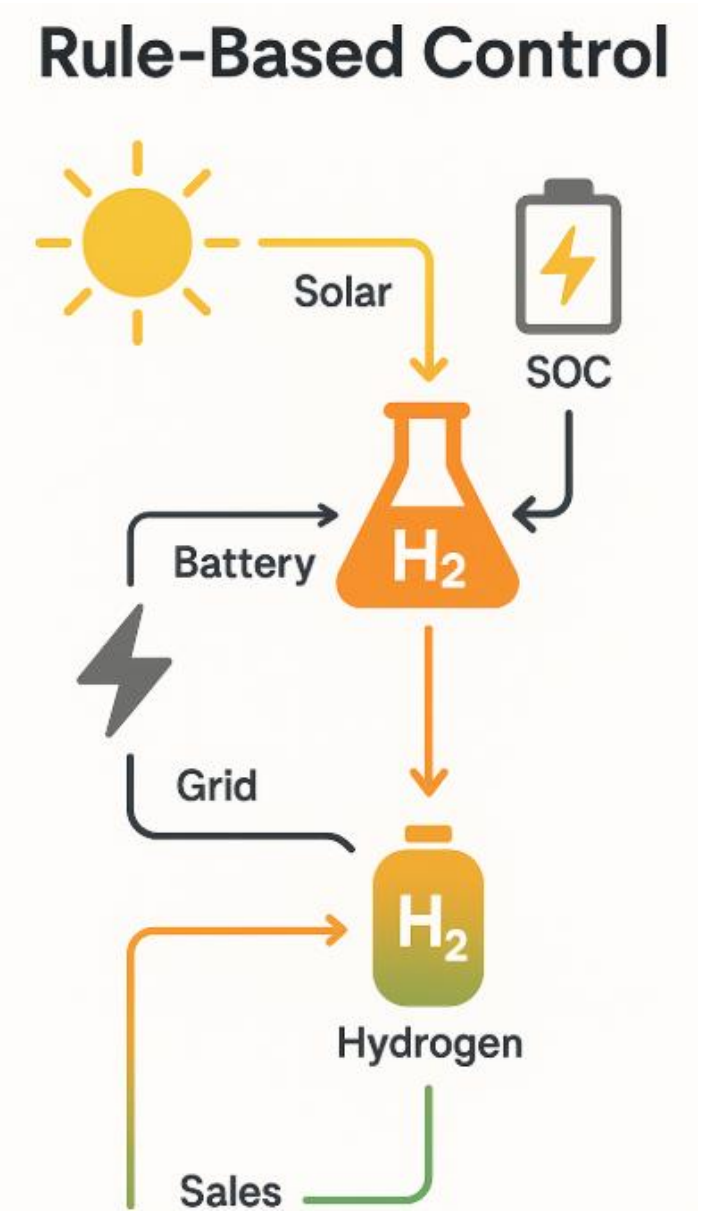
- Data-driven and model-based, more rigorous approach
- Incorporate constraints
- Strategic and tactical planning for large, complex supply chains



2. Rule-based control approach

Rule-Based Control Logic (If, else logic)

- **Solar Usage Priority**
  - Use solar first for the electrolyzer
  - Excess solar → battery if SOC < 70%
- **Battery Charging**
  - Charge battery only when SOC < 80%
  - Stop charging if SOC ≥ 80% or battery limit reached
- **Battery Discharging**
  - Discharge if SOC > 20% and solar is insufficient
  - Only discharge to help meet unmet hydrogen demand
- **Grid Use**
  - Use grid *only* if solar + battery can't meet demand
  - Limit grid power to electrolyzer max − solar − battery
- **Electrolyzer Operation**
  - Run at max using available power (priority: solar → battery → grid)
  - H_2 → meet demand first, then store excess
- **Hydrogen Storage Logic**
  - Store only if storage < 90%
  - Discharge storage only if production < demand

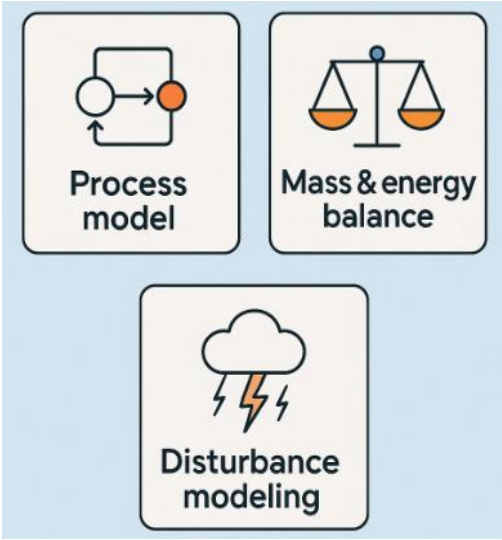


3. MPC workflow components

MPC workflow components

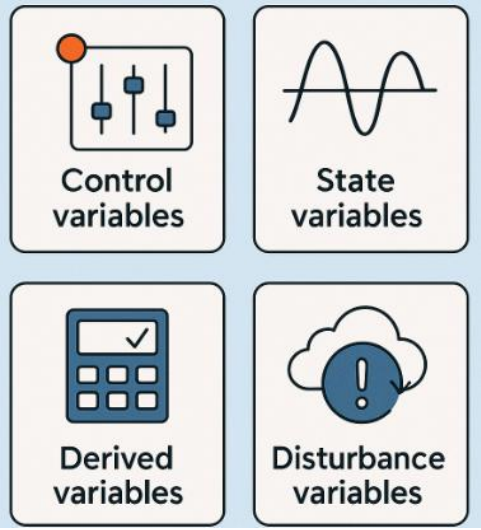
1. System modeling:

- process model, mass & energy balance
- disturbance modeling



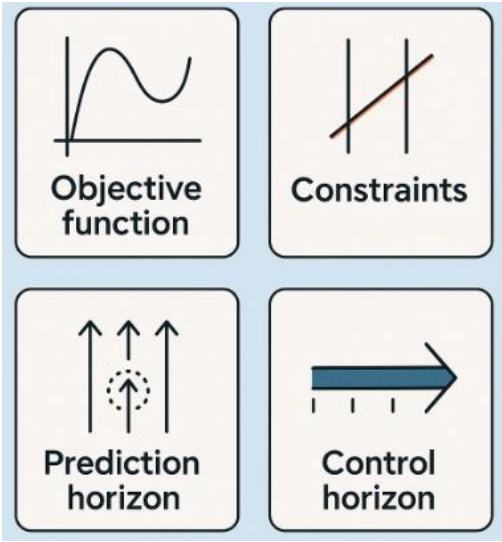
2. System variables:

- control, derived and state variables
- disturbance variables and measured outputs



3. Optimization formulation:

- objective function and constraints
- prediction and control horizons



Process model, mass & energy balance, feasibility & limits

Solar Energy Allocation

$$E_{se}(t) = \text{solar_avail}(t) - E_{sb}(t)$$

Electricity Flow into Electrolyzer

$$E_{se}(t) + E_{be}(t) + E_{ge}(t) = P_{elec}(t)$$

Hydrogen Production via Electrolysis

$$H_{prod}(t) = \eta_{elec} \cdot P_{elec}(t)$$

$$H_{prod}(t) = H_{es}(t) + H_{est}(t)$$

Battery Energy Storage Dynamics

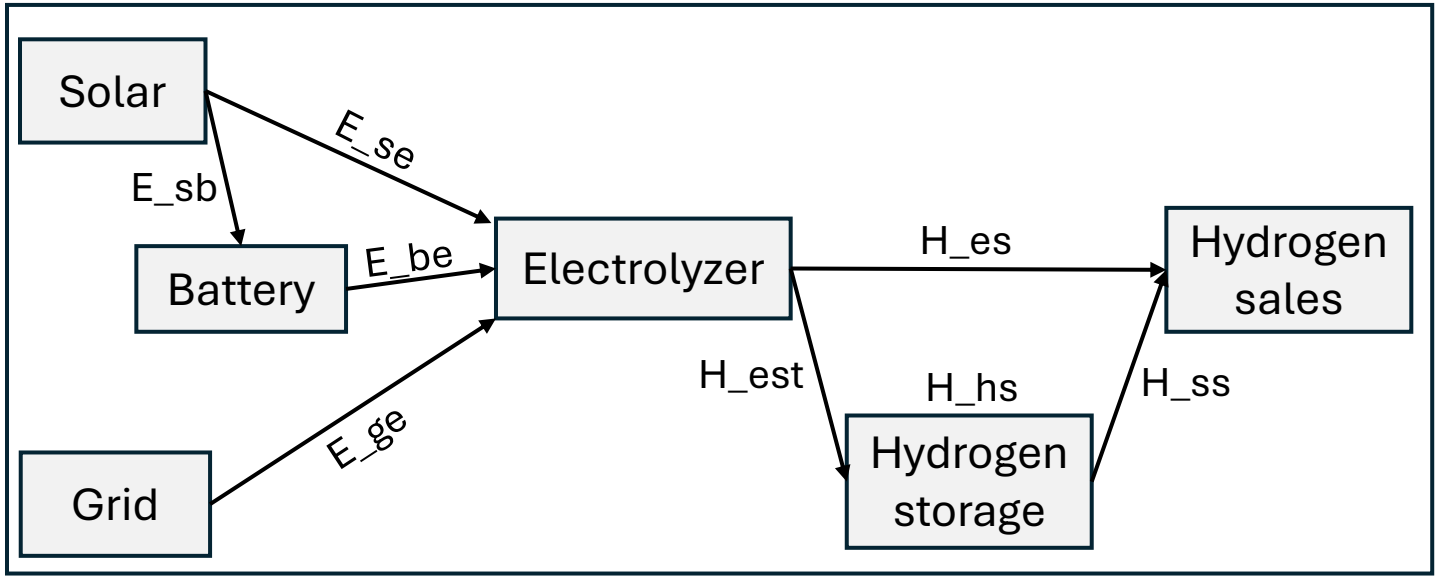
$$SOC(t) = SOC(t-1) + \eta_{ch} \cdot E_{sb}(t) - \frac{E_{be}(t)}{\eta_{dch}}$$

Hydrogen Storage Dynamics

$$H_{hs}(t) = H_{hs}(t-1) + H_{est}(t) - H_{ss}(t)$$

Hydrogen Sales Flow

$$H_{sales}(t) = H_{es}(t) + H_{ss}(t)$$



Feasibility & Limits

- Electrolyzer to sales constraint

$$H_{es}(t) \leq H_{prod}(t)$$

- Storage to sales constraint

$$H_{ss}(t) \leq H_{hs}(t-1)$$

- Total hydrogen sales cannot exceed demand

$$H_{sales}(t) \leq H_{demand}(t)$$

- Storage capacity limit

$$H_{hs}(t) \leq H_{hs}^{\max}$$

- Battery power and SOC constraints

$$E_{sb}(t),\ E_{be}(t) \leq P_{\text{bat}}^{\max}, \quad SOC(t) \leq SOC^{\max}$$

- Electrolyzer power constraint

$$P_{elec}(t) \leq P_{elec}^{\max}$$

Non-Negativity Constraints

$$E_{sb}(t),\ E_{be}(t),\ E_{ge}(t),\ E_{se}(t),\ P_{elec}(t) \geq 0$$

$$H_{prod}(t),\ H_{es}(t),\ H_{est}(t),\ H_{ss}(t),\ H_{sales}(t),\ H_{hs}(t) \geq 0$$

$$SOC(t) \geq 0$$

Disturbance modeling

Real-Time Models (True System)

Solar Availability

$$\text{Solar}_{\text{real}}(t) = \underbrace{100}_{\text{Scale}} \cdot \left(\underbrace{0.8}_{\text{Base level}} + \underbrace{0.1 \cdot \sin\left(\frac{2\pi t}{24}\right)}_{\text{Diurnal fluctuation}} \right)$$

Grid Electricity Price

$$\text{GridPrice}_{\text{real}}(t) = \underbrace{75}_{\text{Base level}} + \underbrace{10 \cdot \cos\left(\frac{2\pi t}{24}\right)}_{\text{Daily fluctuation}}$$

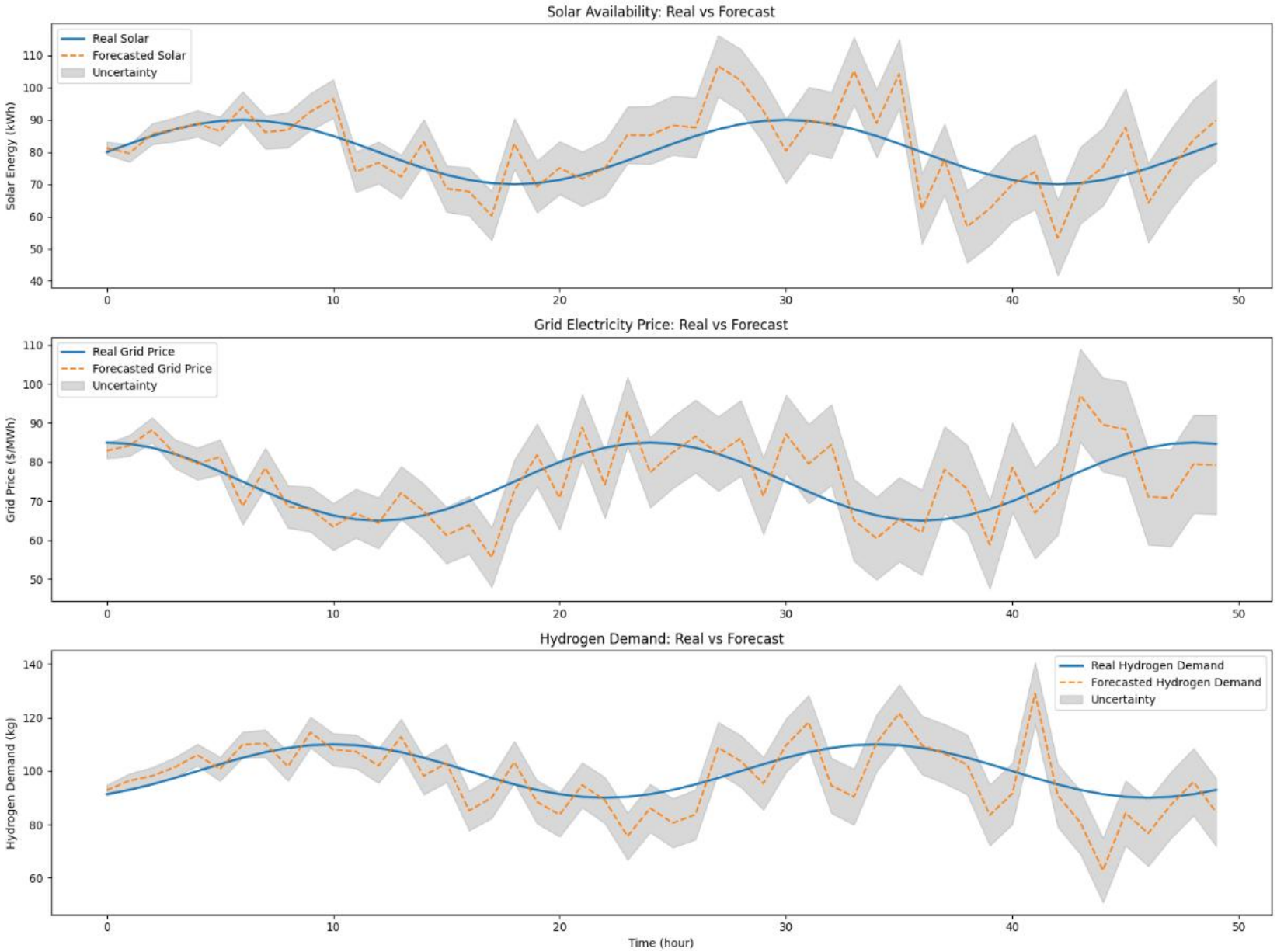
Hydrogen Demand

$$\text{HydrogenDemand}_{\text{real}}(t) = \underbrace{100}_{\text{Base level}} + \underbrace{10 \cdot \sin\left(\frac{2\pi t}{24} - \frac{\pi}{3}\right)}_{\text{Diurnal fluctuation with phase shift}}$$

General Forecast Form

$$\text{Forecast}_X(t) = \text{Real}_X(t) + \epsilon_X(t)$$

$$\epsilon_X(t) \sim \mathcal{N}\left(0,\ \sigma_0^2 \cdot (1 + \alpha t)\right)$$



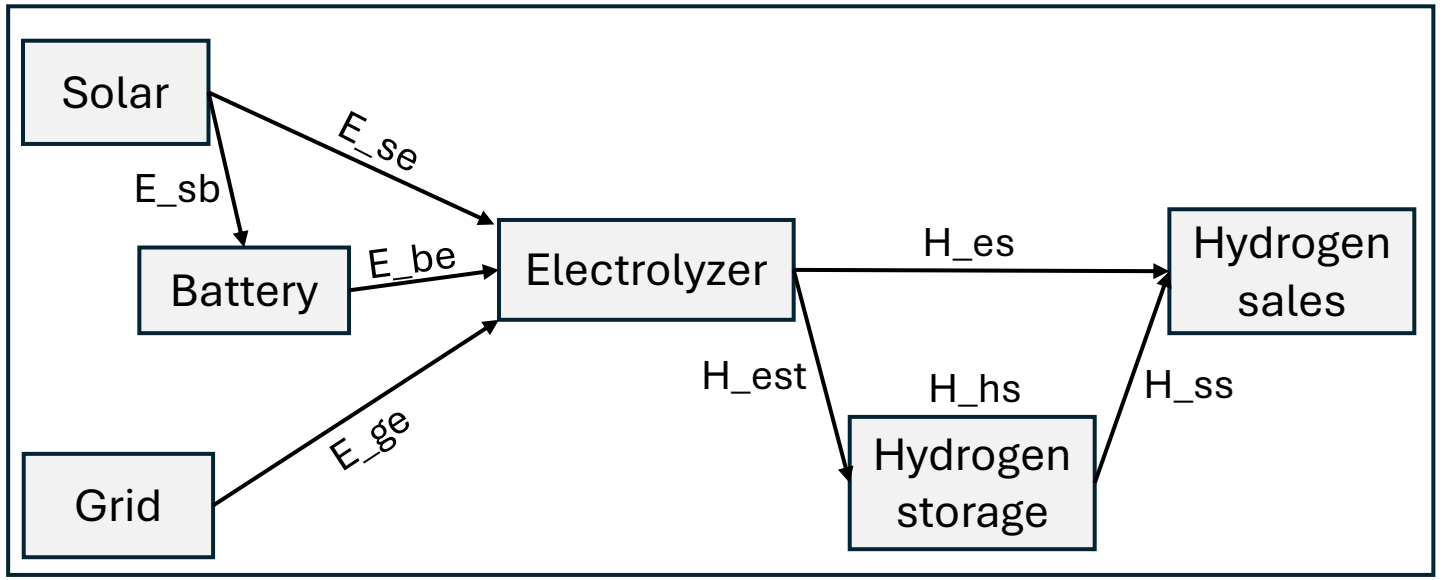
System variables

🔧 Control Variables (Decision Variables)

Symbol	Description
$E_{sb}(t)$	Solar → Battery charging
$E_{be}(t)$	Battery → Electrolyzer
$E_{ge}(t)$	Grid → Electrolyzer
$H_{es}(t)$	Electrolyzer → Sales (direct H_2 sale)
$H_{est}(t)$	Electrolyzer → Storage (H_2 stored)
$H_{ss}(t)$	Storage → Sales (withdrawal from H_2 storage)

📐 Derived Variables

Symbol	Description
$E_{se}(t)$	Solar → Electrolyzer = solar available – $E_{sb}(t)$
$P_{\text{elec}}(t)$	Total electrolyzer input = $E_{se} + E_{be} + E_{ge}$
$H_{prod}(t)$	Hydrogen produced = $\eta_{elec} \cdot P_{\text{elec}}(t)$
$H_{sales}(t)$	Total H_2 sold = $H_{es}(t) + H_{ss}(t)$



📦 State Variables (Tracked over Time)

Symbol	Description
$\text{SOC}(t)$	Battery State of Charge
$H_{hs}(t)$	Hydrogen in storage

🌪 Disturbance Variables (External Inputs)

Symbol	Description
$\text{Solar}_{\text{real}}(t)$	Real solar availability
$\text{GridPrice}_{\text{real}}(t)$	Real grid electricity price
$\text{H}_{\text{demand}}(t)$	Real hydrogen demand

Optimization formulation

OPEX-Only Objective Function for MPC

$$\min \sum_{t=0}^{T-1} \Big[\underbrace{C_{\text{grid}}(t) \cdot E_{ge}(t)}_{\text{grid electricity cost}} + \underbrace{C_{\text{battery}} \cdot (E_{sb}(t) + E_{be}(t))}_{\text{battery throughput O\&M}} + \underbrace{C_{\text{elec}} \cdot P_{\text{elec}}(t)}_{\text{electrolyzer OPEX}} + \underbrace{C_{\text{Hstore}} \cdot H_{hs}(t)}_{\text{H}_2\text{ storage OPEX}} - \underbrace{P_{\text{H2}} \cdot H_{\text{sales}}(t)}_{\text{revenue from H}_2\text{ sales}} \Big]$$

Constraints

Battery SOC Dynamics

$$SOC(t) = SOC(t-1) + \eta_{\text{ch}} \cdot E_{sb}(t) - \frac{E_{be}(t)}{\eta_{\text{dch}}}$$

$$0 \le SOC(t) \le SOC_{\max}$$

Battery Power Limits

$$0 \le E_{sb}(t),\ E_{be}(t) \le P_{\text{bat}}^{\max}$$

Power Balance to Electrolyzer

$$E_{se}(t) = \text{Solar}_{\text{avail}}(t) - E_{sb}(t)$$

$$P_{\text{elec}}(t) = E_{se}(t) + E_{be}(t) + E_{ge}(t)$$

$$P_{\text{elec}}(t) \le P_{\text{elec}}^{\max}$$

Hydrogen Production

$$H_{\text{prod}}(t) = \eta_{\text{elec}} \cdot P_{\text{elec}}(t)$$

$$H_{\text{prod}}(t) = H_{es}(t) + H_{est}(t)$$

Hydrogen Storage Dynamics

$$H_{hs}(t) = H_{hs}(t-1) + H_{est}(t) - H_{ss}(t)$$

$$0 \le H_{hs}(t) \le H_{hs}^{\max}$$

Hydrogen Sales Constraints

$$H_{\text{sales}}(t) = H_{es}(t) + H_{ss}(t)$$

$$H_{\text{sales}}(t) \le H_{\text{demand}}(t)$$

$$H_{es}(t) \le H_{\text{prod}}(t), \quad H_{ss}(t) \le H_{hs}(t-1)$$

Non-Negativity

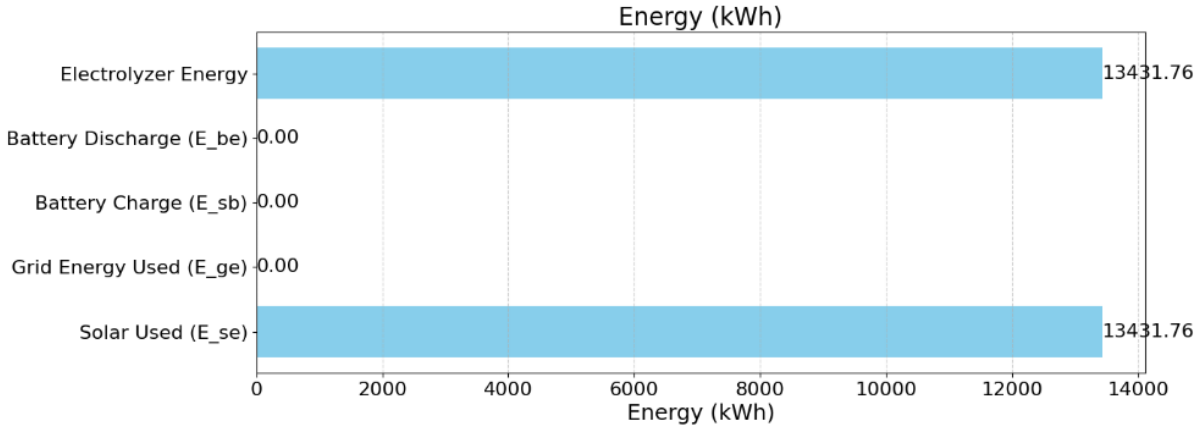
$$E_{sb}(t), E_{be}(t), E_{ge}(t), P_{\text{elec}}(t), H_{\text{prod}}(t), H_{es}(t), H_{est}(t), H_{ss}(t), H_{\text{sales}}(t), H_{hs}(t), SOC(t) \ge 0$$

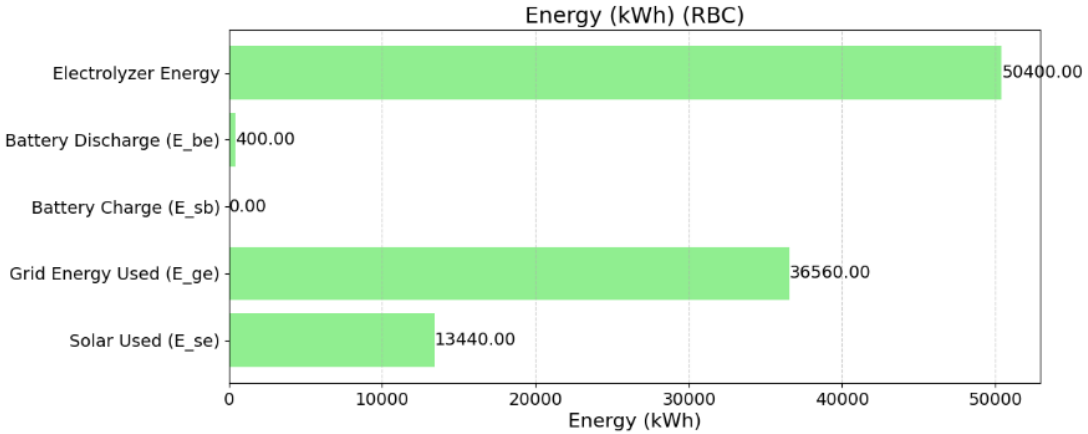
Horizon Settings

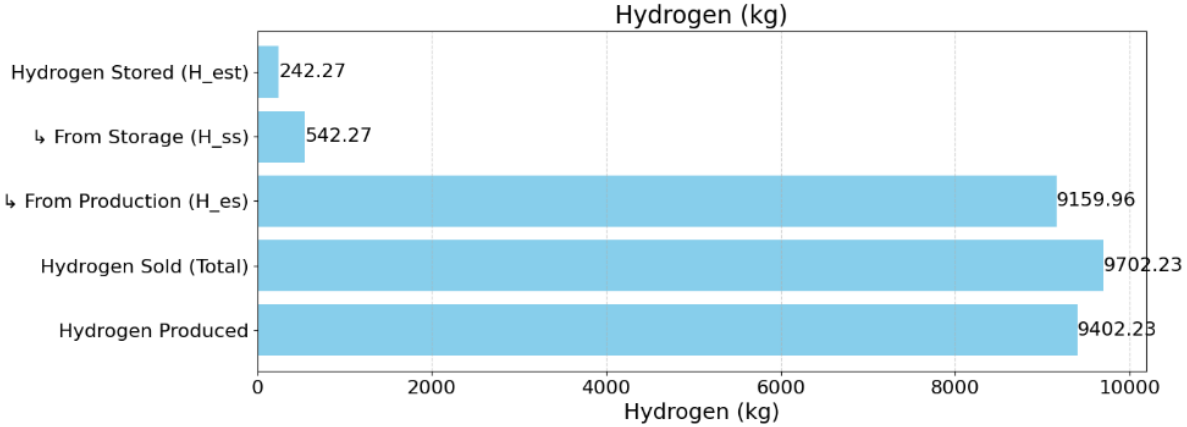
Setting	Value
Prediction Horizon	12 hrs (future steps)
Control Horizon	1 hr (apply 1st step)
Total Duration	168 hrs (1 week)
MPC Solves	168 (1/hr)
Control Steps Used	168
Vars per Solve	72 (6 controls × 12 hr)
Total Vars/Week	12,096 (72 × 168)

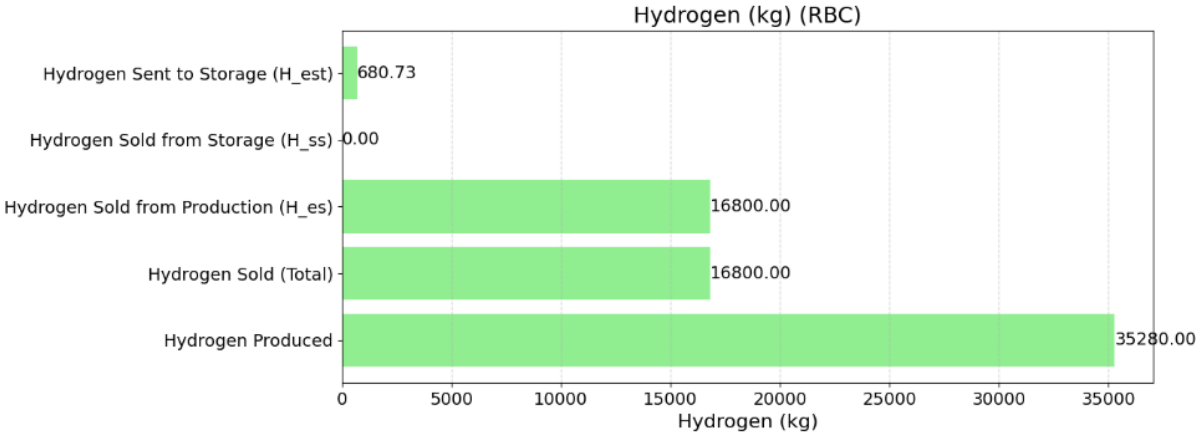
4. Results

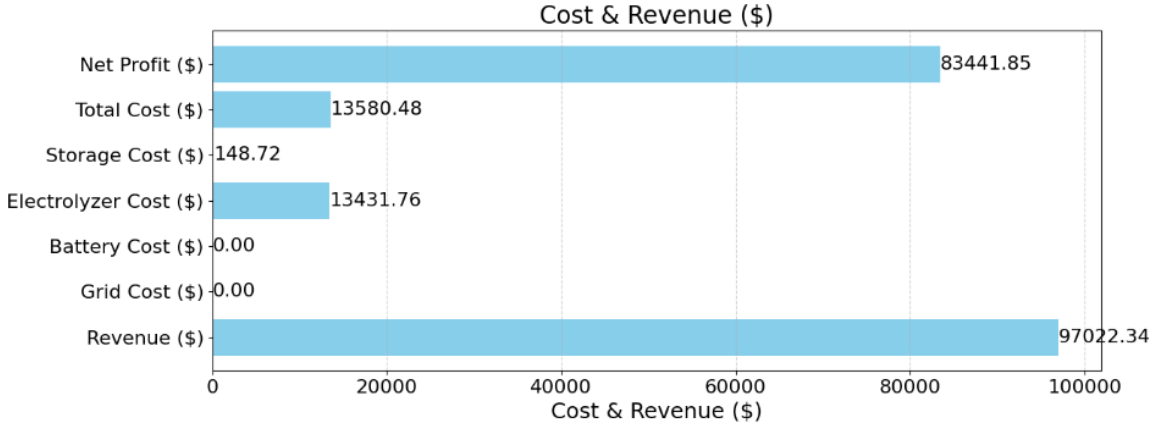
MPC results 2:

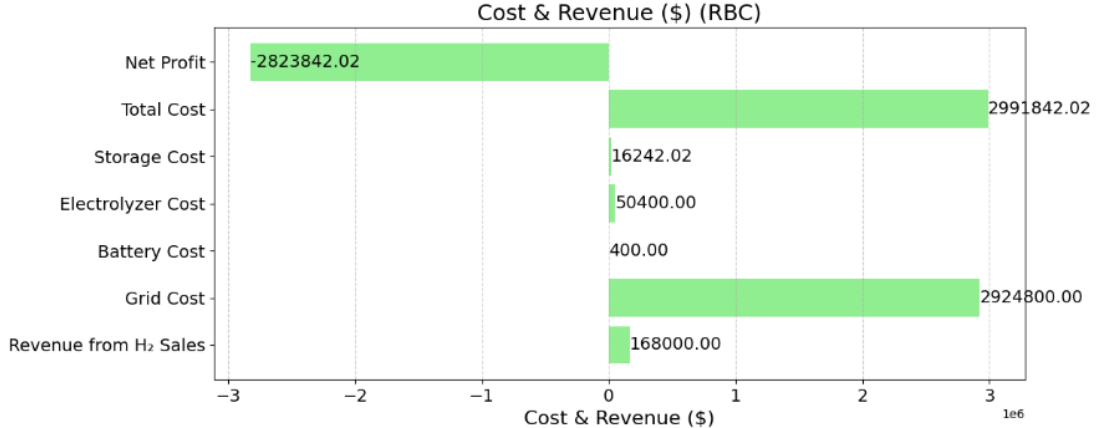












Conclusion and Future work

Conclusions

- **MPC** achieves **high cost-efficiency**, selling hydrogen **without** relying on costly **grid power**
- **RBC** suffers from **overproduction** and **poor utilization**, due to heuristic, reactive logic
- **MPC's forecasting** and **optimization** enable smarter energy allocation
- **RBC overproduces** hydrogen, **wasting** energy and leading to **low system efficiency**
- **Battery** usage remains **low** in both
- **MPC** supports **economically sustainable** operations, leveraging solar while minimizing waste

Future work

- Explore **reinforcement learning** for hydrogen supply chain optimization, offering **adaptive**, **data-driven** control beyond fixed rules and forecast-dependent MPC
- Investigate **multi-agent RL frameworks** to enable **decentralized coordination** and improved **scalability**

Thanks for attention